

Karthik Reddy Duddukunta

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Research Profile

Data Science researcher specializing in deep learning, computer vision, causal inference, and multimodal NLP, with applications in environmental monitoring, sustainability analytics, and infrastructure science. Experienced in designing novel neural architectures, formulating and testing research hypotheses, conducting ablation studies, and rigorously evaluating models on real-world, noisy datasets. Author of peer-reviewed journal and conference publications spanning environmental analytics, healthcare AI, and predictive modeling. Seeking Research Scientist / PhD / MS (Research) roles at the intersection of AI and environmental science.

Education

Master of Science in Data Science Expected 2027

Montclair State University, New Jersey, USA

Research Focus: Deep Learning for Environmental Monitoring, Multimodal AI Systems

Bachelor of Technology in Computer Science and Engineering 2020 – 2024

SRM Institute of Science and Technology, Chennai, India

GPA: 9.56 / 10 (Top 5%)

Technical & Research Skills

Machine Learning & Deep Learning: CNNs, U-Net, FCN, EfficientNet, Transformers, Attention Mechanisms, Multimodal Learning, LLMs, RAG

Frameworks & Libraries: PyTorch, TensorFlow, Scikit-learn, spaCy

Data & Analytics: Python (Pandas, NumPy, Matplotlib), SQL, Statistical Modeling, Exploratory Data Analysis

NLP & Knowledge Systems: Topic Modeling (BERTopic, LDA), NER, Text Classification, Knowledge Graphs (Neo4j)

Research Tools: Git, LaTeX, Jupyter, Google Colab

Domain Expertise: Environmental AI, Microplastic Analysis, Scientific Literature Mining, Sustainability Analytics, Healthcare AI

Research Experience

Research Assistant May 2025 – Present

Clean Energy and Sustainability Analytics Center (CESAC), Montclair State University

- Led applied machine learning research for sustainability and clean energy analytics, formulating research questions on climate-driven energy patterns to inform institutional planning.
- Developed and validated predictive statistical and ML models (including survival analysis) on 5,000+ real-world records to support workforce and sustainability decision-making.
- Conducted exploratory and inferential analysis on large institutional datasets, applying hypothesis testing to identify long-term climate and energy trends.
- Mentored five master's research teams on experimental design, data pipelines, model evaluation, and scientific communication.
- Collaborated with interdisciplinary researchers to translate data-driven findings into peer-reviewed, policy-relevant insights.

ElectricAI — Causal & Forecasting Analysis of Energy Costs May 2026 – Present

Causal Inference, Time-Series Forecasting, Agentic RAG — DoWhy, EconML, XGBoost, Prophet, SARIMAX, LangGraph, LangChain

- Investigated the causal relationship between electricity pricing and consumption behavior using DoWhy and EconML Double Machine Learning, isolating price-elasticity effects from weather confounders (HDD/CDD).
- Designed a comparative time-series forecasting study of energy demand across XGBoost, Prophet, and SARIMAX to benchmark model families under varying seasonal and regional conditions.
- Developed an agentic Retrieval-Augmented Generation (RAG) architecture (LangGraph, LangChain) as a research prototype for automated evidence retrieval and claim validation in energy-domain question answering.
- Independently built the underlying full-stack research platform (React/TypeScript, FastAPI) to support experimentation, benchmarking, and analysis.

Utility Pole & Wire Detection — Geometric Characterization from Street-Level Imagery 2025

YOLOv8-OBB, DINO, SAM, Active Learning, Tiled Inference

- Developed a research pipeline for utility/electric pole detection and geometric characterization from high-resolution street-level imagery using YOLOv8 Oriented Bounding Boxes (OBB), achieving **mAP@50 = 0.95**, **Precision = 0.92**, **Recall = 0.90**.
- Estimated pole **height** and **leaning angle** from OBB geometry and image metadata, enabling automated structural risk screening for infrastructure-monitoring applications.
- Designed an automated annotation and quality-control workflow integrating foundation-model object discovery (DINO/SAM) to reduce manual labeling effort and scale dataset construction.
- Investigated detection failure modes in high-resolution mobile imagery and developed an active learning framework with tiled/sliced inference to improve robustness for small, distant objects.

RESEARCH PROJECTS

EfficientNetV2-Based Stage-Adaptive Attention Network for Microplastic Segmentation

- Proposed a stage-adaptive attention encoder-decoder architecture using EfficientNetV2-B3 to address extreme class imbalance and scale variability in microscopy images.
- Integrated multi-resolution feature fusion and resolution-aware attention to improve boundary delineation and semantic discrimination.
- Achieved **mIoU of 0.8248** and **Dice coefficient of 0.7932**, outperforming existing microplastic segmentation approaches.
- Enabled automated environmental microplastic monitoring and scalable pollution analysis.

Publications

Journal Articles

- Shrestha, K., Duddukunta, K. R., Lal, P., Hasan, M. M., Rahman, M. M., Pradhan, N. S. (2026). *Compound Multi-Hazard Assessment under CMIP6 Climate Scenarios: Tracking Seasonal Flood-Landslide and Drought-Fire Interactions Across Nepal*. *Natural Hazards*, 122(14), 542. [\[Google Scholar\]](#)

Conference Papers

- Praba, M. S. B., Duddukunta, K. R., Bezawada, V. S., Addanki, S. V. (2024). *Enhancing Web Security Through Machine Learning: A Random Forest Approach to Malicious URL Detection*. 2024 4th Asian Conference on Innovation in Technology (ASIANCON), IEEE, 1–6.
- Praba, M. S. B., Duddukunta, K. R., Bezawada, V. S., Addanki, S. V. (2024). *Predicting Real Estate Price: A Comparative Analysis of Linear Regression and Support Vector Machine*. Congress on Control, Robotics, and Mechatronics, Springer Nature Singapore, 123–134.

Preprints

- Somkuwar, P., Duddukunta, K. R., Anu, V., Lal, P., Chivukula, U., Sharma, P., Shrestha, K. (2026). *Speak Safe: An AI System for Robust Voice Authentication with Deepfake Detection and Noise Reduction*. ResearchSquare (preprint). [\[Google Scholar\]](#)

Research Service & Leadership

Hackathon Judge, TRAVEL-WIZARD: Designing Conversational Travel AI (2024)

Conference Representative, NJLM Conference (2025)

Research Mentor, CESAC (2025–Present)

Awards & Certifications

- Northeast Big Data Innovation Hub – Explorer Transportation Data Science Project (2024)
- NPTEL – Python for Data Science (Elite)
- NPTEL – Database Management Systems (Elite)
- Coursera – Data Analysis and Visualization with Power BI